Anti-Prompt Injection Defense

Security and Privacy

- Blocks Jailbreaks: detect and stop attempts to trick the model into breaking rules.
- Sanitizes Inputs: Neutralizes hidden commands or malicious instructions in user prompts.
- Protects System Prompts: Ensures the core identity and instructions of the AI remain intact.
- Prevents Manipulation: Stops users from overriding safety protocols via clever phrasing.

Hallucination Prevention & Fact-Checking

Response and Relevance

- Verifies Claims: Real-time cross-referencing of generated facts with trusted knowledge bases.
- Reduces Confabulations: Filters out plausible-sounding but incorrect information.
- Cites Sources: Ensures every claim is backed by a verifiable reference or document.
- Grounds Responses: Forces the model to use provided context rather than making things up.

Sensitive Data Screening (PII)

Security and Privacy

- Redacts Personal Info: Automatically masks names, phones, emails, and addresses.
- Prevents Leakage: Stops the model from inadvertently revealing training data secrets.
- Complies with Privacy Laws: Helps adhere to GDPR, CCPA, and data protection standards.
- Filters Financial Data: Blocks credit card numbers and banking details from outputs.

Language Toxicity Detection

Security and Privacy

- Filters Hate Speech: Blocks slurs, discriminatory remarks, and harassment.
- Monitors Tone: Ensures interactions remain polite and professional.
- Detects Aggression: Identifies and neutralizes hostile or threatening language.
- Safe for Work: Guarantees content is appropriate for professional environments.

Contextual Relevance Check

Response and Relevance

- Stays on Topic: Ensures the answer directly addresses the user's specific question.
- Filters Noise: Removes tangential or unrelated information from the response.
- Validates Coherence: Checks if the response logically follows the conversation history.
- Prevents Drift: Stops the model from wandering into unrelated subjects.

SQL Injection Prevention

Logic and Functionality Validation

- Sanitizes Queries: Detects and blocks malicious SQL code in natural language inputs.
- Validates Syntax: Ensures generated SQL is syntactically correct and safe to run.
- Restricts Scope: Prevents unauthorized database access or modification attempts.
- Protects Data Integrity: Shields your database from destructive commands.

Competitor Shield & Brand Safety

Content Validation and Integrity

- Blocks Competitor Mentions: Prevents the AI from promoting or discussing rival products.
- Enforces Brand Tone: Ensures responses align with your company's voice and values.
- Filters Negative Sentiment: protecting your brand reputation in generated text.
- Controls Recommendations: strictly suggests your own services and solutions.

Link Validity Checker

Response and Relevance

- Tests URLs: Pings generated links to ensure they are live and not 404s.
- Prevents Dead Ends: Removes or replaces broken links before showing them to users.
- Verifies Safety: Checks linked sites against malicious domain lists.
- Ensures Accessibility: Guarantees resources shared are actually reachable.

Intent Alignment Verification

Response and Relevance

- Analyzes Goal: Confirms the response actually solves the user's underlying problem.
- Checks Instruction Following: Verifies if constraints (e.g 'answer in 3 bullets') were met.
- Prevents Misunderstandings: Clarifies ambiguous queries before answering.
- Ensures Satisfaction: Aligns output format and depth wit user expectations.

Structured Data/JSON Validator

Logic and Functionality Validation

- Enforces Schema: Checks if generated JSON matches the required structure/keys.
- Fixes Syntax: Auto-corrects missing braces, quotes, or commas.
- Type Checking: Ensures values (strings, ints) matched expected data types.
- Parsability Guarantee: Ensures downstream systems ca read the AI's output without crashing.

Multilingual Accuracy Audit

Language Quality

- Preserves Nuance: Ensures translations keep the origina meaning and tone.
- Checks Grammar: Validates syntax and vocabulary in th target language.
- Cultural Context: Adapts idioms and references for the specific audience.
- Prevents Mistranslation: Flags potential errors in critical communications.

Coherence & Logic Filter

Content Validation and Integrity

- Detects Gibberish: Filters out repetitive or nonsensical text loops.
- Checks Argument Flow: Ensures premises logically lead to conclusions.
- Removes Hallucinated Words: Spots non-existent words or garbled syntax.
- Quality Assurance: Guarantees a polished, human-readable output.

Complexity & Readability Scoring

Language Quality

- Adjusts Level: Measures Flesch-Kincaid score to match audience expertise.
- Simplifies Jargon: Explains technical terms for general audiences.
- Enhances Clarity: Refines sentence structure for maximum readability.
- Tone Matching: Switches between formal/technical and casual/simple styles.

Dynamic Pricing Verification

Content Validation and Integrity

- Real-Time Lookup: Validates quoted prices against current database values.
- Prevents Hallucinated Discounts: Stops the AI from inventing fake deals.
- Currency Check: Ensures correct currency symbols and formatting.
- Inventory Sync: Only quotes prices for items actually in stock.

API Schema Compliance

Logic and Functionality Validation

- Validates Rest Calls: Checks if generated API requests match the OpenAPI spec.
- Parameter Check: Ensures required fields are present and valid.
- Method Enforcement: Verifies correct HTTP methods (GET, POST, etc.) are used.
- Endpoint Accuracy: Prevents calls to non-existent or deprecated API endpoints.

Citation & Source Tracing

Content Validation and Integrity

- Attribution Check: Ensures quotes are correctly attributed to their authors.
- Link Verification: Connects claims directly to the source document.
- Prevents Plagiarism: Rewrites content that is too similar to training data.
- Audit Trail: Provides a clear path of where information came from.

Redundancy Removal

Language Quality

- De-duplicates Ideas: Merges repetitive points into concise statements.
- Trims Fluff: Removes filler words that add no value.
- Conciseness Optimization: Shortens content without losing meaning.
- Streamlines Output: Creates punchy, direct responses.

Output Quality Assessment

Language Quality

- Grades Relevance: Scores how well the answer fits the prompt.
- Evaluates Completeness: distinct score for missing information.
- Feedback Loop: Uses low scores to trigger retries or human review.
- Quality Standard: Enforces a minimum quality bar before showing the user.

Logical Consistency Checker

Logic and Functionality Validation

- Spot Contradictions: identifying self-conflicting statements in a single response.
- Temporal Consistency: Ensures timelines and dates make chronological sense.
- Entity Consistency: Keeps names and roles consistent throughout the text.
- Argument Integrity: logic flow validation.

Inappropriate Content Filter

Security and Privacy

- NSFW Filter: Strictly blocks adult or explicit content generation.
- Violence Detection: Stops generation of gory or graphically violent text.
- Self-Harm Prevention: Intervenes if user prompts suggest self-harm.
- Safety First: The baseline layer of protection for all interactions.